

ECON 102: INEQUALITY FORMULAS

In this document, we present the most important formulas used in Module 7.

Income Share

To illustrate, suppose the population is composed of 20 individuals with the following incomes:

39	104	76	193	114	134	183	61	75	99
143	103	32	121	111	54	130	117	152	57

The first step is to sort the income:

32	39	54	57	61	75	76	99	103	104
111	114	117	121	130	134	143	152	183	193

Income Share by Quintile

We group the population in five equally populated groups from the poorest to the richest and we compute the total income in each group:

					Total
Quintile 1	32	39	54	57	182
Quintile 2	61	75	76	99	311
Quintile 3	103	104	111	114	432
Quintile 4	117	121	130	134	502
Quintile 5	143	152	183	193	671

Then, we divide the total of each group by the total income. For example, the income share in the first quintile is

$$\frac{182}{182 + 311 + 432 + 502 + 671} \times 100 = 8.6749\%$$

The income shares by quintile in percentage are therefore:

Quintile 1	8.6749
Quintile 2	14.8236
Quintile 3	20.5910
Quintile 4	23.9276
Quintile 5	31.9828

Cumulative Income Share by Quintile

The cumulative sum of a list of numbers $\{X_1, \dots, X_n\}$ is the following list:

$$\begin{aligned} &X_1 \\ &X_1 + X_2 \\ &X_1 + X_2 + X_3 \\ &X_1 + X_2 + X_3 + X_4 \\ &\vdots \\ &X_1 + X_2 + X_3 + X_4 + \dots + X_n \end{aligned}$$

Apply this definition to the income shares and we obtain the cumulative income shares. The second column is the cumulative population share (20% of population are in each quintile).

	Cum. income share	Population Share
Quintile 1	8.7	20.0
Quintile 2	23.5	40.0
Quintile 3	44.1	60.0
Quintile 4	68.0	80.0
Quintile 5	100.0	100.0

Gini Coefficient From Quintiles

Let h_i be the cumulative income share for quintile i not expressed in percentage, then the Gini coefficient is:

$$G = 0.8 - 0.4 \times (h_1 + h_2 + h_3 + h_4)$$

In this case, it is equal to 0.2228.

Income Share by Decile

We group the population in 10 equally populated groups from the poorest to the richest and we compute the total income in each group:

			Total
Decile 1	32	39	71
Decile 2	54	57	111
Decile 3	61	75	136
Decile 4	76	99	175
Decile 5	103	104	207
Decile 6	111	114	225
Decile 7	117	121	238
Decile 8	130	134	264
Decile 9	143	152	295
Decile 10	183	193	376

Then, we divide the total of each group by the total income as for the quintile case. The income shares by decile in percentage are therefore:

Decile 1	3.3842
Decile 2	5.2908
Decile 3	6.4824
Decile 4	8.3413
Decile 5	9.8665
Decile 6	10.7245
Decile 7	11.3441
Decile 8	12.5834
Decile 9	14.0610
Decile 10	17.9218

Cumulative Income Share by Decile

We apply the same formula presented earlier for the quintiles:

	Cum. income share	Population Share
Decile 1	3.4	10.0
Decile 2	8.7	20.0
Decile 3	15.2	30.0
Decile 4	23.5	40.0
Decile 5	33.4	50.0
Decile 6	44.1	60.0
Decile 7	55.4	70.0
Decile 8	68.0	80.0
Decile 9	82.1	90.0
Decile 10	100.0	100.0

Gini Coefficient From Quintiles

Let h_i be the cumulative income share for decile i , then the Gini coefficient is:

$$G = 0.9 - 0.2 \times (h_1 + h_2 + h_3 + h_4 + h_5 + h_6 + h_7 + h_8 + h_9)$$

In this case, it is equal to 0.2324.

The Median

The median is the number that separates a sample in two equal groups after sorting the observations.

Odd Number of Observations

If the series is $\{X_1, X_2, X_3, X_4, X_5\}$ with $X_1 < X_2 < X_3 < X_4 < X_5$, the median is equal to X_3 .

Even Number of Observations

If the series is $\{X_1, X_2, X_3, X_4, X_5, X_6\}$ with $X_1 < X_2 < X_3 < X_4 < X_5 < X_6$, the median (one version of it) is equal to $(X_3 + X_4)/2$.